

- How many elements does the set $\{W, E, L, O, V, E, S, E, T, T, H, E, O, R, Y\}$ have?
- Which of these sets are the same? Are any subsets of others?
 $A = \{a, s, l, e, e, p\}$ $B = \{e, a, s, e, l, s\}$ $C = \{s, e, p, a, l\}$
 $D = \{r, e, e, l, s\}$ $E = \{p, e, a, r, l, s\}$ $F = \{r, e, l, a, p, s, e\}$
 $G = \{l, a, s, e, r\}$ $H = \{l, e, s, s, e, r\}$ $I = \{l, e, a, s, e\}$
- Let $\mathcal{E}$ be the universal set of lowercase letters in the English alphabet. Consider the following subsets:
 $A = \{w, e, l, c, o, m, e\}$, $B = \{t, o, t, h, e\}$, $C = \{s, i, x, t, h, f, o, r, m\}$.
 How many elements do each of the following sets have?
 a. $A \cap B$ b. $A \cap C$
 c. $B \cup C$ d. $A \cap C'$
- With universal set $\mathcal{E} = \mathbb{N}$, let $T = \{2x : x \in \mathbb{N}\}$, $S = \{x^2 : x \in \mathbb{N}\}$, and P be the set of (positive) prime numbers. Illustrate T , S , and P on a Venn diagram.
- Let Q be the set of planar quadrilaterals. Consider the following subsets of Q :
 - $L = \{x \in Q : x \text{ has at least one line of symmetry}\}$
 - $A = \{x \in Q : x \text{ has at least one pair of equal angles}\}$
 - $S = \{x \in Q : x \text{ has at least one pair of parallel sides}\}$
 - $D = \{x \in Q : \text{the diagonals of } x \text{ meet at right angles}\}$
 - $R = \{x \in Q : x \text{ is a rhombus}\}$

What do the following statements say in 'plain English'. Are they true? If the statement is not true, then provide a counterexample?

- $L \subseteq A$
 - $L \cap S \subseteq R$
 - $S \cap D \subseteq R$
 - $L' \subseteq R'$
 - $R' \subseteq L'$
 - $A \cap S = \emptyset$
 - $R' \cap D \subseteq S'$
 - $A' \subseteq S' \cap D'$
 - $A \subseteq L \cup S$
- With the universal set $\mathcal{E}$ of positive integers less than or equal to 100, let T be the set of multiples of 10 and S be the set of multiples of 6.
 - Draw a Venn diagram showing S and T , adding the elements of $S \cup T$ in their correct places (you need *not* show the elements of $\mathcal{E}$ which are not in either set).
 - What is the significance of the smallest element of $S \cap T$? What is the simplest way of describing the elements of $S \cap T$?
 - With the universal set $\mathcal{E}$ of triangles, let E be the set of equilateral triangles, R be the set of right angled triangles and I be the set of isosceles triangles.
 - Draw a Venn diagram showing the relationship between E , R and I .
 - Add to your diagram, in the correct places, the triangles a , with sides 3, 4 and 5; b , with sides 3, 5 and 5; c with sides 3, 5 and 6 and d with sides 4, 4 and 4.
 - Give the side lengths of a triangle that is in $R \cap I$. How are all the triangles in $R \cap I$ related to each other?
 - Scalene triangles are those with all three side lengths different. Write a statement in set notation to show how the set S of scalene triangles is related to the sets already named in this question?
 - On separate Venn diagrams shade the sets
 - $A \cap B$
 - $A' \cup B'$
 - $A \cup B$
 - $A' \cap B'$

What do you observe? Explain in English why these results are obvious, now you've seen them.

9. Which of the following statements are:

- always true,
- can be true or false, or
- never true?

Justify your answers with proofs and counterexamples.

- | | |
|-------------------------------------|---|
| a. $(A' \cup B) \cap A = \emptyset$ | b. $(A' \cap B) \not\subseteq (A \cup B')'$ |
| c. $(A \cup B)' \subseteq A'$ | d. $(A' \cap B') = (A \cap B)'$ |
| e. $(A \cap B)' \subseteq B'$ | |

10. Let $\mathcal{E} = \{\text{plane shapes}\}$, $M_n = \{\text{shapes with exactly } n \text{ mirror lines}\}$, and $R_n = \left\{\text{shapes that look the same when they are rotated by an angle of } \frac{360}{n}^\circ\right\}$.

- a. Explain why $M_2 \cap M_6 = \emptyset$.
- b. Explain why $R_6 \subseteq R_2$.
- c. Investigate true claims that can be made about these sets. For instance, is $M_2 \subseteq R_2$?